

220 V Siren Circuit

The receiver closes a relay when the water passes the danger mark. The relay switches the siren's mains supply.

Read this before you build anything

220 V mains kills. The low-voltage half of this circuit is a beginner project; the mains half is not. Have the mains wiring done or inspected by a qualified electrician, work only with the supply physically disconnected, and never build mains wiring on a breadboard or with jumper wires. If this is for a competition demo, consider using a 12 V siren and a battery instead — it demonstrates exactly the same thing and nothing in this document can hurt you.

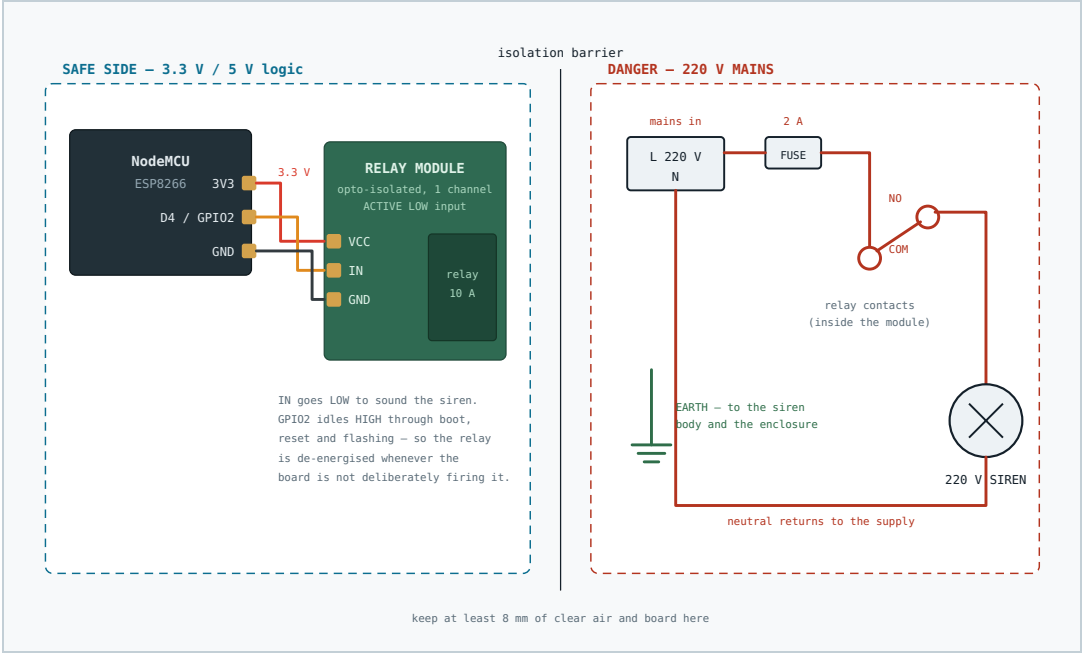


Fig. 1 The relay is the only thing that crosses the barrier, and it crosses it as an air gap inside a sealed component. No wire runs between the two halves.

Low-voltage connections

NODEMCU	RELAY MODULE	NOTE
3V3	VCC	Use a module rated for 3.3 V logic, or feed its logic side 3.3 V and its coil side 5 V with the JD-VCC jumper removed
GND	GND	Common ground
D4 / GPIO2	IN	Active low — see below

WHY THE INPUT MUST BE ACTIVE LOW

GPIO2 is a strapping pin: the board holds it HIGH at power-up, through every reset, and while you flash new firmware. An active-low module reads that as "off", so the siren stays silent whenever the ESP8266 is not deliberately firing it. Wire an active-high module to the same pin and it will scream every time you press reset — and, worse, whenever the board crashes.

GPIO2 WAS THE ONBOARD LED

The sketch no longer drives it. The external LED on D2 still shows the level, and the dashboard still shows everything. If you want an onboard indicator back, you have no free pins left on this wiring — drop DIO0 from D1 (the code polls, it does not use the interrupt) and move the LED there.

Mains side, and how to build it safely

The rules that matter

1. **Switch the live conductor, not the neutral.** If you switch neutral, the siren stays at mains potential even when silent.
2. **Fuse the live side** before the relay — 2 A is ample for a siren. A relay that welds its contacts shut is a fire, not an inconvenience.
3. **Earth the siren body and any metal enclosure.** Green-yellow conductor, star point, no daisy chains.
4. **Use the normally-open contact.** A power cut then leaves the siren silent rather than stuck on.
5. **Keep 8 mm of clear air and board** between the mains tracks and everything else. No jumper wires crossing that gap, ever.
6. **Use proper mains cable and screw terminals.** Dupont jumpers are rated for neither the voltage nor the current.
7. **Put the whole thing in an insulated enclosure** with strain relief on the cable entries, and run the supply through an RCD.
8. **Test with a table lamp first.** When the lamp switches on and off correctly at the right water level, swap in the siren.

PART	SPEC	WHY
Relay module	1-ch, opto-isolated, 10 A 250 VAC	Derate hard: a 10 A relay is right for a 1 A siren
Fuse + holder	2 A, in the live line	Protects against a welded contact or a shorted siren
Mains cable	0.75 mm ² three-core	Live, neutral, earth
Enclosure	Insulated, IP54 or better	Keeps fingers out and water off
Siren	220 V AC, under 2 A	Check the rating before you rely on the relay

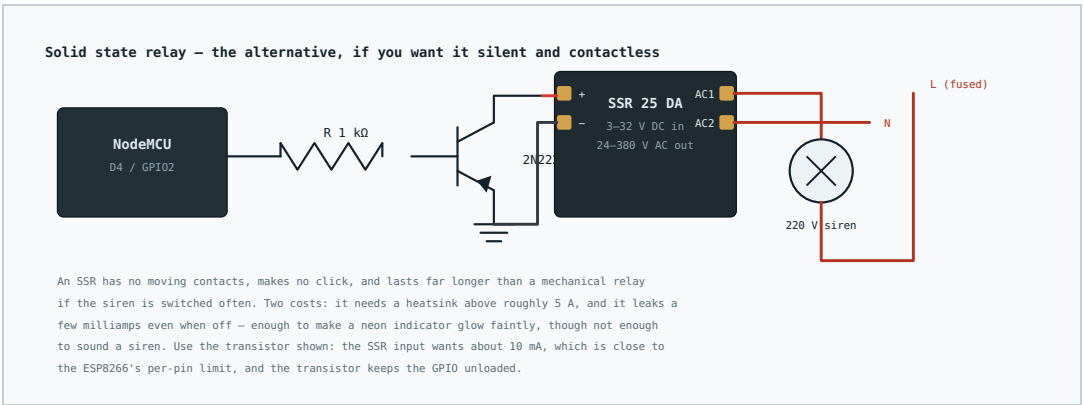


Fig. 2 A solid state relay is the alternative. Same firmware, same pin, no moving parts.

How the firmware behaves

SITUATION	SIREN	WHY
Water below the danger mark	silent	—
Water passes the mark	sounds	The alert flag from the sender
Water drops back, under 30 s later	keeps sounding	Minimum on-time, so it cannot chatter on a wave
Radio link dies while in alert	keeps sounding	Losing the radio during a flood is when you need it most
Someone presses Silence on the page	stops	Re-arms automatically once the water is confirmed safe
ESP8266 resets, crashes, or is re-flashed	silent	GPIO2 idles high — the relay simply de-energises

That fourth row is the one worth arguing about. The LED version cleared its alert when packets stopped, because a stale light is misleading. A siren is different: it must not fall silent because the radio failed. So the alert latches, the dashboard shows **STALE — LINK LOST**, and the siren keeps sounding until someone silences it by hand or the water is confirmed safe.

```
#define PIN_SIREN      2          // D4 / GPIO2, drives an ACTIVE-LOW relay module
#define SIREN_MIN_MS  30000UL   // once it sounds, it sounds for 30 s minimum

void sirenWrite(bool on) { digitalWrite(PIN_SIREN, on ? LOW : HIGH); }

// in setup(), as the very first thing that runs:
pinMode(PIN_SIREN, OUTPUT);
sirenWrite(false);

// in loop():
stale = (goodCount > 0) && (now - lastHeard > 6000);

if (alert && !silenced && !sirenOn) {
  sirenOn = true; sirenSince = now;
  Serial.println("*** SIREN ON - level past the danger mark ***");
}
if (sirenOn) {
  bool minElapsed = (now - sirenSince) > SIREN_MIN_MS;
  if (silenced || (!alert && !stale && minElapsed)) sirenOn = false;
}
if (!alert && !stale) silenced = false;    // re-arm once the water drops
sirenWrite(sirenOn);
```

While the siren is sounding, a red SILENCE SIREN button appears next to the status pill at 192.168.4.1. It calls /silence on the receiver. The siren re-arms by itself once the level is back below the mark.